ZYVORAI LABS · CUSTOMER DOCUMENTATION

Chimera

Page-by-Page Product Manual

2026-08-29

Page-by-page guides

Each guide follows: Purpose → When to use it → How to get there → Operate from the console (UX) → Related pages.

Every route is also listed in the complete page index.

Auth

Page	What it covers
Login	Authenticate the browser session before Command Center chrome or live data loads.

Command Center

Page	What it covers
Top Activity	Donut chart and legend grouping infrastructure traffic by operation class — vSphere SOAP, SDK, NFC download, NFC resume, and related gateway paths.
Fault Studio	Operator drawer for latency, HTTP failure status, next-N API/NFC failures, one-shot NFC stream drops, drop offset, and bandwidth caps.
Live requests	Stream recent infrastructure HTTP calls: method, compact path, HTTP status, and end-to-end gateway duration.
Login	Authenticate the browser session before any Command Center chrome or data is shown.
Overview KPIs	Show live gateway-derived health: requests, error rate, active sessions, bytes transferred, NFC exports, and average response latency.
Scenario Launcher	One-click deterministic presets for clean, slow, flaky, and resume-export fault paths — without hand-tuning Fault Studio each run.
Settings	Show the configured listen address (read-only) and change Command Center admin username/password live — no Chimera restart required.
Topology	Visual datacenter → cluster → host → datastore map of the simulated estate.
VM inventory	Searchable, paginated table of simulator VMs with power state, CPU/memory/disk, datastore, and Fixture badge (real VMDK vs synthetic).
VMDK Library	List and manage `.vmdk` fixtures under configured fixture directories; upload from the browser or pick staged host files; optionally pin to a VM.

---

11 guides. Regenerate: `node scripts/customer-docs/generate-guide-index.mjs`.

Login

Purpose

Authenticate the browser session before Command Center chrome or live data loads.

When to use it

- Open this surface when the job matches the purpose above
- Start from the product home / dashboard if you are unsure where to begin
- Confirm auth and that required backends/operators are reachable if data looks empty

How to get there

- Route / id: `/login`
- Nav: **AUTH → Login** (sidebar, command palette, or desktop nav)

What you can do

1. Open `/login` and wait for live data from Chimera.
2. Use filters and search when the page provides them.
3. Drill into a row or card for detail, then jump to related surfaces.
4. For mutating actions: review impact, role gates, and confirmation dialogs first.

If the page stays empty, check service health, auth configuration, and that dependencies for this domain are installed.

Related pages

- Getting Started
- Page index

Top Activity

Purpose

Donut chart and legend grouping infrastructure traffic by operation class — vSphere SOAP, SDK, NFC download, NFC resume, and related gateway paths.

When to use it

- See *what kind* of traffic dominates during Transiva export or discovery runs
- Confirm NFC vs SOAP mix after resume/retry scenarios
- Quick sanity check when KPIs move but you need operation-level breakdown

How to get there

- Screen id: `#activity`
- Nav: **Control Center → Telemetry** (sidebar ∿) or scroll to **Top Activity** on Overview

Operate from the console (UX)

1. Sign in and scroll to the **Top Activity** card below the topology panel.
2. Read the center total — it tracks overall request count from telemetry (same family as the Requests KPI).
3. Inspect colored legend rows: each line shows operation name, count, and percentage share.
4. Run `./bin/chimera selftest` or a Transiva export, then watch segments populate within ~2s polling cycles.
5. If the donut is a flat ring with **No traffic yet.**, drive clients against `/sdk` — dashboard fetches do not count.

Empty / fail: No traffic yet. persists during active exports → verify Transiva `vcenter_url` points at `<host>:8989/sdk`, not localhost from a remote runner. Legend shows one class at 100% unexpectedly → check Fault Studio for forced failures skewing retries.

Success: Multiple operation classes appear with plausible percentages; total matches rising **Requests** KPI during a self-test or export.

Related pages

- Overview KPIs
- Live requests
- Scenario Launcher
- Workflows
- Page index

Fault Studio

Purpose

Operator drawer for latency, HTTP failure status, next-N API/NFC failures, one-shot NFC stream drops, drop offset, and bandwidth caps.

When to use it

- Fine-grained fault injection beyond Scenario Launcher presets
- Hardening Transiva or other clients against partial transfers and API errors
- Inspecting and clearing active fault policy before clean benchmark runs

How to get there

- Drawer id: `#faultDrawer` (not a route — slides over the console)
- Nav: **Administration → Fault Studio** (sidebar ϟ), top bar **Fault Studio** (bolt icon), or **View Fault Studio →** on Scenario Launcher
- **Settings** (gear icon) is a separate drawer — do not confuse with Fault Studio

Operate from the console (UX)

1. Sign in — mutating fault APIs require the admin session token.
2. Open Fault Studio from sidebar, top bar bolt, or scenario panel link.
3. Read summary chips at top: **Latency**, **Bandwidth**, **Status** (current policy snapshot).
4. Set fields as needed:
    - **Latency (ms)** — artificial delay on infrastructure requests.
    - **Failure HTTP status** — status returned for armed failures (default 503).
    - **Fail next requests** — count of generic API failures before auto-clear.
    - **Fail next NFC requests** — NFC-specific failure budget.
    - **Drop next NFC streams** / **Drop after (MiB)** — mid-transfer disconnect for resume testing.
    - **Bandwidth cap (MiB/s, 0 = unlimited)** — throttle NFC throughput.
5. Click **Apply policy** — toast **Fault policy active**; watch Live requests for red statuses.
6. Click **Reset** for a clean traffic path without reloading the page (same as Clean scenario family).
7. **Unlock** / **Lock** toggles admin token in this tab; **API Tokens** sidebar entry uses the same control.
8. Close drawer with **×** or click the dimmed overlay.

Empty / fail: Drawer opens but **Apply** sends you to login → re-authenticate. Changes seem ignored → confirm toast success; some counters are consume-on-use (next-N). Bandwidth shows **∞** when cap is 0.

Success: Summary chips match inputs; injected errors appear in live request feed; **Reset** restores green statuses and nominal latency.

Related pages

- Scenario Launcher
- Live requests
- Overview KPIs

- Workflows
- Admin basics
- Page index

Live requests

Purpose

Stream recent infrastructure HTTP calls: method, compact path, HTTP status, and end-to-end gateway duration.

When to use it

- Debug govmomi / Transiva traffic and spot failing calls (red status codes)
- Correlate Fault Studio or scenario changes with immediate API/NFC errors
- Audit-style tail of the newest simulator-facing requests (also linked as **Sessions / Audit Logs** in the sidebar)

How to get there

- Screen id: `#requests`
- Nav: **Control Center → Sessions** or **Audit Logs** (sidebar), or **Live Requests** panel on Overview
- Shortcut: top bar bell (badge = current error count) scrolls directly here

Operate from the console (UX)

1. Sign in and open **Live Requests** (or click the bell when the badge is non-zero).
2. Watch rows populate: **GET/POST** method chip, truncated path, green status (or red for ≥400).
3. Compare duration column (ms) before/after **Slow Fabric** scenario or Fault Studio latency.
4. Click **View All** to expand from 10 to 30 buffered rows (label flips to **View Less**); no extra API call — same server buffer.
5. When debugging resume exports, look for NFC paths returning **206** after **Resume Export** scenario.
6. Cross-check with Fault Studio when statuses match your injected `fail_status`.

Empty / fail: Waiting for client traffic… during an export → client may be pointed at wrong host/port or TLS mismatch. All red rows → run **Clean Environment** scenario, then **Reset** in Fault Studio. Bell badge stuck high after clean run → click ↻ **Refresh** on Overview.

Success: SOAP/SDK/NFC paths appear with 2xx/206 statuses on happy path; error badge clears after clean reset.

Related pages

- Fault Studio
- Overview KPIs
- Top Activity
- Workflows
- Page index

Login

Purpose

Authenticate the browser session before any Command Center chrome or data is shown.

When to use it

- Every new browser tab or after **Log out** from the top-right user menu
- Before Scenario Launcher, Fault Studio writes, VMDK upload, or credential changes

How to get there

- URL: `http(s)://<host>:8989/__chimera/`
- Screen id: login gate (full page — no sidebar until authenticated)

Operate from the console (UX)

1. Open `http(s)://<host>:8989/__chimera/` in a browser.
2. Confirm you see **Log in to Chimera** — not the overview metrics (unauthenticated users never flash inventory).
3. Enter username and password (default **admin** / **admin** unless changed in Settings or via `CHIMERA_ADMIN_*`).
4. Click **Log in** (or press Enter in the password field).
5. Wait for the toast **Logged in — Scenarios and fault injection are enabled.** The sidebar, KPI row, and inventory load.

Empty / fail: Login failed toast → verify credentials in Settings (if you still have another session) or env vars on the host. Reload loops usually mean wrong password or stale token — use **Log out** from the user chip, then sign in again.

Success: Full Command Center shell appears; sidebar health ring shows uptime/version; footer shows active vSphere persona and SDK endpoint.

Console: `http(s)://<host>:8989/__chimera/` · vSphere SDK: `http(s)://<host>:8989/sdk` · Never publish lab IPs — use `<host>`.

Related pages

- Settings
- Overview KPIs
- Admin basics
- Getting Started
- Page index

Overview KPIs

Purpose

Show live gateway-derived health: requests, error rate, active sessions, bytes transferred, NFC exports, and average response latency.

When to use it

- First screen after login — confirm Chimera is receiving infrastructure traffic
- Before and after scenario or fault changes to watch error rate and latency move
- When Transiva or `selftest` runs but you need a quick "is anything happening?" check

How to get there

- Screen id: `#overview` (Command Center home)
- Nav: **Control Center → Overview** (sidebar ⌂) or land here immediately after login

Operate from the console (UX)

1. Sign in at `/__chimera/` if the login gate is showing.
2. Read the six metric tiles: **Requests**, **Error Rate**, **Active Sessions**, **Data Transfer**, **Exports**, **Avg Response**.
3. Note trend arrows (↑/↓) on the second poll — dashboard polling under `/__chimera` is excluded from these counters, so flat zeros mean no client traffic yet.
4. Optionally change **Last 15 minutes** / **Last hour** in the page head (visual filter; telemetry still polls every ~2s).
5. Click ↻ **Refresh** to force an immediate telemetry pull if you just started Transiva or `selftest`.
6. Use the top bar **Scenario** dropdown shortcut to scroll to Scenario Launcher when you want presets without hunting the page.

Empty / fail: All KPIs stay at zero after export tests → confirm clients hit `/sdk`, not `/__chimera/`. **Error Rate** climbs with red statuses in Live requests — open Fault Studio or run **Clean Environment**. Health ring in the sidebar reads **Unavailable** → check `GET /__chimera/health` on the host.

Success: Requests and sessions increment during `selftest` or Transiva export; exports tick up on NFC lease starts; error rate stays near 0% on clean runs.

Related pages

- Live requests
- Top Activity
- Topology
- Scenario Launcher
- Getting Started
- Page index

Scenario Launcher

Purpose

One-click deterministic presets for clean, slow, flaky, and resume-export fault paths — without hand-tuning Fault Studio each run.

When to use it

- Repeatable Transiva or client hardening tests
- Reset to baseline (**Clean Environment**) between scenarios
- Simulate NFC drop mid-transfer for Range/206 resume behavior

How to get there

- Screen id: `#scenarios`
- Nav: **Control Center → Scenarios** (sidebar ◈) or top bar **Scenario** button (scrolls here)

Operate from the console (UX)

1. Sign in — scenarios POST to `/__chimera/scenario/*` and require the admin bearer token.
2. Scroll to **Scenario Launcher** (right column on Overview, below VM inventory on wide layouts).
3. Click a card:
   - **Clean Environment** — reset faults and counters (happy path baseline).
   - **Slow Fabric** — high latency and low bandwidth pressure.
   - **Flaky API** — random failures and timeouts on API calls.
   - **Resume Export** — drop NFC connection mid-transfer (~2 MiB) for Range retry / HTTP 206.
4. Active card highlights with an inset ring; toast shows **Scenario applied** with the preset name.
5. Open Fault Studio to inspect the underlying latency/failure fields the preset set.
6. Click **View Fault Studio →** at the bottom of the card for the same drawer.
7. Re-run Transiva export or `./bin/chimera selftest`, then watch Live requests and KPI error rate.

CLI equivalent (same token as API): `./scripts/scenario.sh clean|slow|flaky|resume`

Empty / fail: Click does nothing / login page returns → session expired; sign in again. **Scenario failed** toast → check engine logs; verify `Authorization` path with `CHIMERA_ADMIN_TOKEN` if using curl. Resume test never shows 206 → confirm client sends Range headers (see Workflows).

Success: Toast confirms preset; Fault Studio summary chips update; traffic reflects scenario (latency, errors, or dropped NFC) until **Clean Environment** or **Reset**.

Related pages

- Fault Studio
- Live requests
- Overview KPIs
- Workflows
- Page index

Settings

Purpose

Show the configured listen address (read-only) and change Command Center admin username/password live — no Chimera restart required.

When to use it

- After first login in a shared lab — rotate off default **admin / admin**
- Confirm which address/port Chimera binds (`CHIMERA_LISTEN`) before sharing URLs
- Same panel as **Users & Auth** — Chimera has one shared admin login, not a multi-user directory

How to get there

- Drawer id: `#settingsDrawer`
- Nav: **Administration → Settings** or **Users & Auth** (sidebar), or top bar **gear** icon

Operate from the console (UX)

1. Sign in (required to save credential changes).
2. Open Settings from sidebar **Settings**, **Users & Auth**, or the top bar gear — all three open this drawer.
3. Read **Listen address** at top — reflects live bind (for example `0.0.0.0:8989`); changing it needs `CHIMERA_LISTEN` in env/config and a process restart.
4. Enter **New admin username** and **New admin password** (both required).
5. Click **Save login** — toast **Login updated**; fields clear.
6. Log out via the top-right user chip and sign in with the new credentials on next session.
7. Optional: set `CHIMERA_ADMIN_USERNAME` / `CHIMERA_ADMIN_PASSWORD` at startup for immutable deploys — Settings overrides persist in memory until restart.

Empty / fail: Missing fields toast → supply both username and password. **Save failed** while logged out → login gate appears; authenticate first. Listen shows **—** → bootstrap could not read bind metadata; check config file.

Success: New credentials work on next login; old password rejected; listen address matches what you use in `http(s)://<host>:8989/__chimera/` .

Related pages

- Login
- Fault Studio (separate drawer — bolt icon, not gear)
- Admin basics
- Getting Started
- Page index

Topology

Purpose

Visual datacenter → cluster → host → datastore map of the simulated estate.

When to use it

- Orient operators to inventory shape before export or Transiva wiring
- Confirm config counts (`datacenters` , `clusters` , `hosts_per_cluster` , `datastores`) match expectations
- Walk new teammates through the fake vSphere layout without opening the SDK

How to get there

- Screen id: `#topology`
- Nav: **Control Center → Topologies** (sidebar ⌘) or scroll to **Infrastructure Topology** on Overview

Operate from the console (UX)

1. From Overview, locate the **Infrastructure Topology** card — subtitle shows live counts (for example `1 datacenter · 2 clusters · 3 hosts`).
2. Read node labels: **DC0** (datacenter), **Cluster0/1**, **Host0–2** with lab IPs, **Datastore0–2** with fixture size from config.
3. Click **View Full Topology** to expand the same diagram full-viewport; use toolbar **Fit** to reset zoom or to step zoom in.
4. Dismiss fullscreen via **Exit Full Screen**, click outside the diagram, or press **Escape**.
5. Cross-check host/datastore count with VM inventory sidebar badge and table footer.

Empty / fail: Subtitle stuck at generic text → bootstrap API may have failed; reload after login or check engine logs. Nodes show but datastore labels say `fixture` with wrong size → verify `fixture_size_mb` in config and restart if you changed estate shape.

Success: Topology subtitle matches your JSON config; fullscreen zoom works; footer **vSphere Persona** strip shows the public SDK endpoint clients should use.

Related pages

- VM inventory
- Overview KPIs
- Admin basics
- Getting Started
- Page index

VM inventory

Purpose

Searchable, paginated table of simulator VMs with power state, CPU/memory/disk, datastore, and Fixture badge (real VMDK vs synthetic).

When to use it

- Pick an export target for Transiva or manual `ExportVm` tests
- Confirm fixture matching (`Matched`, `Round-robin`, `Shared`, `Manual`, `Synthetic`)
- Filter powered-on VMs before copying a name to the clipboard

How to get there

- Screen id: `#inventory`
- Nav: **Control Center → Inventory** (sidebar ▤) — badge shows total VM count
- Global search (**⌘K** / Ctrl+K) filters this table and scrolls to it

Operate from the console (UX)

1. Sign in and scroll to **Virtual Machines (*n*)** (or use sidebar **Inventory**).
2. Use **Search VM by name...** or the top-bar global search — both filter the same table.
3. Set **All Power States** to **Powered On** or **Powered Off** when narrowing export candidates.
4. Scan columns: vCPU, memory, disk, datastore, **Fixture** badge (hover for underlying filename).
5. Click the ⇩ icon on a row to copy that VM name to the clipboard (toast confirms selection).
6. Click ⎔ **Export VM** in the card header to copy the first row matching current filters — useful after power-state filter.
7. Paginate with numbered page buttons (5 VMs per page); sidebar count should match config `vms_per_pool` totals.

Empty / fail: No matching virtual machines. with empty search → check simulator counts in config and restart Chimera after estate changes. All **Synthetic** when you expected real disks → stage VMDKs in VMDK Library or set `fixture_vmdk_dir`. Copy actions toast **No VM selected** → relax filters.

Success: Expected VM names (for example `DC0_C0_RP0_VM0`) listed; fixture badges reflect matching rules; copied name pastes into Transiva config or shell scripts.

Related pages

- VMDK Library
- Topology
- Workflows
- Getting Started
- Page index

VMDK Library

Purpose

List and manage `.vmdk` fixtures under configured fixture directories; upload from the browser or pick staged host files; optionally pin to a VM.

When to use it

- Exercise Transiva **conversion** with real disks, not only synthetic NFC transfer
- Inspect which VM (if any) each on-disk fixture is assigned to
- Add fixtures without shell access to the host (upload or browse within allowed roots only)

How to get there

- Screen id: `#vmdks`
- Nav: **Control Center → VMDK Library** (sidebar ⛁) — badge shows file count

Operate from the console (UX)

1. Sign in and scroll to **VMDK Library (*n*)** — subtitle shows the active directory path or **No fixture_vmdk_dir configured**.
2. Review the table: file name, size, assigned VM, assignment method (`Matched` , `Round-robin` , `Manual` , etc.).
3. Click ⇪ **Upload VMDK** (requires login):
    - **Upload a file:** choose a local `.vmdk` , optionally pick **Assign to VM** (or leave **Auto** for name-match / round-robin).
    - Click **Upload** — toast confirms; library rescans within ~5s even without restart.
4. Or **pick a file already on the host:** select fixture root from dropdown, navigate folders (↑ goes up one level), click a `.vmdk` row — you must pick a VM first when assigning from browse.
5. After manual pin, confirm the VM row in VM inventory shows **Manual** fixture badge.
6. Files with no VM assignment still appear here — they will not show in the VM-keyed inventory table until matched.

Empty / fail: No VMDK directory configured. → set `fixture_vmdk_dir` / `fixture_vmdk_dirs` in config and restart. **Upload failed** → check disk space and 8 GiB upload cap. Browse list empty → only configured fixture roots are visible (never arbitrary host paths). **Assign failed** → select target VM before clicking a file in browse mode.

Success: New files appear with size and method; assigned VM column populated; Transiva export of that VM uses the real disk.

Related pages

- VM inventory
- Workflows
- Product repo `docs/TRANSIVA.md` (matching rules)
- Admin basics
- Page index

Chimera — Complete page index

Every primary navigable dashboard route.

Generated: 2026-08-29 · 10 routes

Regenerate: `node scripts/customer-docs/generate-page-index.mjs`

AUTH

Page	Route	Purpose	Guide
Login	`/login`	Authenticate the browser session before Command Center chrome or live data loads.	Open

COMMAND CENTER

Page	Route	Purpose	Guide
Overview KPIs	`/overview`	Six live KPIs — requests, error rate, sessions, bytes, exports, latency — refreshed every two seconds.	Open
Topology	`/topology`	Datacenter → cluster → host → datastore map of the configured simulator estate.	Open
Top Activity	`/activity`	Donut and legend grouping infrastructure traffic by operation class (SOAP, SDK, NFC).	Open
Live requests	`/live-requests`	Newest infrastructure HTTP calls with method, path, status, and gateway duration.	Open
VM inventory	`/vm-inventory`	Searchable, paginated simulator VM table with power state, sizing, datastore, and fixture badge.	Open
VMDK Library	`/vmdk-library`	Fixture `.vmdk` inventory — upload from browser or pick staged host files; optional VM pin.	Open
Scenario Launcher	`/scenarios`	One-click presets: clean, slow fabric, flaky API, resume export.	Open
Fault Studio	`/fault-studio`	Administrative drawer for latency, status codes, next-N failures, NFC drops, bandwidth cap.	Open
Settings	`/settings`	Read-only listen address plus live admin username/password rotation (no restart).	Open

Related

- Customer docs home
- Page-by-page guides